

Vaishnav Mukund Mankar

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SUMMARY

Computer Engineering student specializing in backend systems, distributed IoT architectures, and performance optimization. Experienced in Rust, Go, Python, and C++ with expertise in low-latency communication bridges (OCPP, ZMQ, MQTT) and high-throughput data pipelines (gRPC, TimescaleDB, Redis). Adept at orchestrating AI coding agents as a force multiplier under direct human system design control.

EDUCATION

Pune Institute of Computer Technology (PICT) <i>B.E. Computer Engineering; CGPA: 9.2</i>	Pune, India 2023 – 2027
Global Public School <i>Class XII (HSC); Score: 80%</i>	Pune, India 2023
Ryan International School <i>Class X (SSC); Score: 90%</i>	Pune, India 2021

EXPERIENCE

Backend & Systems Engineer Intern <i>Galain</i>	May 2026 – Present <i>Remote/Hybrid</i>
- Engineered a high-performance EV proxy system in Rust for Enough Energy (UK) to connect physical charging stations with central software utilizing the OCPP-J 1.6 protocol over WebSockets. - Designed a low-latency messaging loop using ZeroMQ (ZMQ) and MQTT to route real-time telemetry, using an embedded Sled DB cache to prevent telemetry data loss during temporary network drops. - Leveraged AI-agent workflows to generate boilerplate and test suites under direct system design control, accelerating deployment speed while owning database schemas, query optimizations, and container orchestration.	
Freelance Full-Stack Developer & Agent Operator <i>Independent</i>	April 2026 <i>Pune, India</i>
- Built a student performance analytics dashboard for investor presentations, integrating physical pen-tech hardware telemetry (CAN/Modbus sensors) with a preprocessed backend data pipeline using React, TypeScript, and Firebase. - Designed dynamic KPI charts, visualizations, and client-side AI summarizers to provide real-time strengths, weaknesses, and writing dynamics analysis. - Managed AI coding agents to rapidly scaffold UI components and routing, focusing manual cognitive effort on system schema, database design, and hardware-software data mapping.	
AI Developer Intern <i>Banao.tech</i>	Aug 2025 – May 2026 <i>Remote</i>
- Engineered backend components for an AI-driven EdTech video generation pipeline, automating personalized multilingual video creation from images and textual summaries. - Learned and applied python-langchain, GitHub workflows, AWS Lambda serverless execution, and cloud logging telemetry to optimize execution and debug latency.	

PROJECTS

FieldPulse (Industrial IoT Backend Platform)	Tech Stack: Go, gRPC, TimescaleDB, Redis, MQTT, OpenTelemetry (OTLP)
- Developed a distributed industrial IoT telemetry ingestion backend in Go, utilizing gRPC microservices and TimescaleDB hypertables. - Optimized database storage using TimescaleDB time-partitioning, automatic 7-day compression (90% savings), and data retention policies. - Integrated Redis for sensor telemetry deduplication and alert suppression, and instrumented all services with OpenTelemetry to trace across HTTP, gRPC, and MQTT boundaries.	
AI-Powered Smart Photo Gallery (digital-asset-management)	Tech Stack: React, FastAPI, PostgreSQL, Gemini, Docker
Engineered a private, local-first "Google Photos" alternative allowing natural language search queries on unstructured media, implementing AI-driven semantic image tagging and text-to-SQL search query optimizations.	

ACHIEVEMENTS

IEEE Publication: Published "GNN-Based Friend Recommendation: A Graph Embedding Approach for Social Networks" in an IEEE International Conference.

TECHNICAL SKILLS

Languages: Rust, Go, Python, C++, SQL, TypeScript, JavaScript
Frameworks: FastAPI, React, Next.js, LangChain, ZeroMQ, gRPC
Cloud & Tools: AWS, Docker, Git, Linux, TimescaleDB, PostgreSQL, Redis, Firebase, OpenTelemetry (OTLP)
Concepts: Data Structures & Algorithms (DSA), Operating Systems (OS), Database Management Systems (DBMS), Computer Networks (CN), Object-Oriented Programming (OOP), Concurrency, System Design, IoT Protocols (OCPP, MQTT)